

Joule heating for SMA wire

1. Introduction

As I illustrated in my previous documentation that my model was ideal model accepts direct temperature signal with no regard of the real life physics as the sudden jumps of the temperature, this documentation is first to solve one of our two main problems that are mentioned in the main documentation, this documentation contains the electro-thermal equations of the joule heating system to take a step further to show how shape memory alloy wire acts in real life with heating and cooling are existing factors .

2. The configuration of Joule heating system

Before I move to the electro-thermal equations I should illustrate how joule heating works, we have a DC power supply with 12V to heat the wire to certain temperature you should apply the electric power for certain time this electric power is two valued the full power or no power there is no in between power values so the time here is the main criteria to reach that certain temperature you say we have a 1 ohm wire so our electric power is $12 \cdot 12 = 144$ watt you need to deliver this 144 watt through the wire for a given time to reach the desired temperature then after you reach it you need to maintain it that is the time the role of Pulse Width Modulation (PWM) comes , the control input of the PWM is something called duty cycle , in simple way duty cycle if we have 1 second of operating time you control how much time of this 1 second the power is on within , If we have 50% duty cycle you will deliver 144 watt for a half second , also the PWM operates in frequencies the previous example operates at 1 Hz ,so the power can go on and off too many time at one seconds so determining the percentage of the duty cycle is the key here to reach a certain temperature and maintain it as much as the signal goes ,In example you are at 293 K and you need to reach 300 K so the duty cycle will be 100% to reach 300 K and then it goes down to less than 100% just to maintain it if it stays at 100% the temperature will increase if it goes to 0% the temperature will decrease to the room temperature ,that's how the PWM controls the whole system , also we have MOSFET as protection of the circuit in prevents high currents to go through the system, to control the duty cycle we have a PID controller implemented with saturation in range $[0, 1]$ that's it for the control and the protection of the circuit ,for the physics of the Joule heating system I will unfortunately describe it with equations although the simplicity of it I would rather not to stuff any equations of these documentation related to this topic but it is the only way to clarify how does the Joule heating is working

$$\frac{dT}{dt}_{\text{net}} = \left(\frac{V^2}{R \cdot mC_p} \right) - \left(\frac{hA_s(T - T_{\infty})}{mC_p} \right)$$

The equation says the rate of change of the temperature equals the electric power multiplied by the thermal mass reciprocal minus the heat loss multiplied by the thermal mass reciprocal

The power supply voltage (V)	12 V
The wire resistance (R)	1.08 Ohm
The mass of the wire (m)	.25 gram
The specific heat capacity (Cp)	400 J/Kg.K
The convective heat transfer coefficient (h)	515 W/m ² .K
The surface area (As)	1.5708 * 10 ⁻⁴ m ²
The room temperature (T _∞)	293 K

Table (1)The electro-thermal parameters

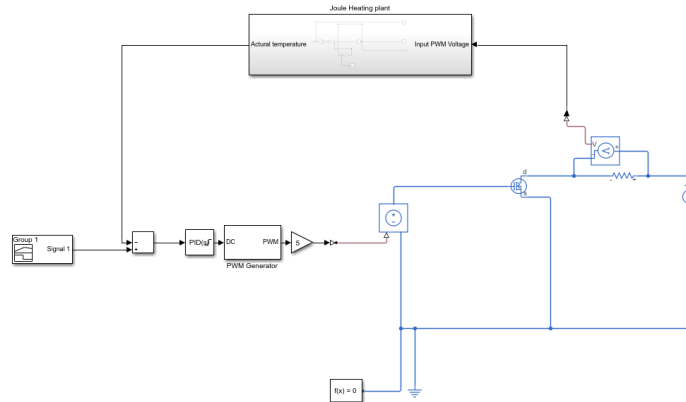


Figure (1) The block diagram of the system.

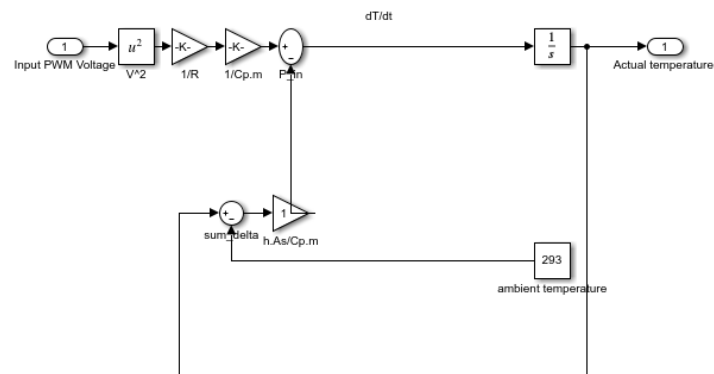


Figure (2) The joule heating system

3. The control of Joule heating

In this section I will discuss the capability of the Joule heating and how it acts in heating step and cooling step and how the convection will affect the system

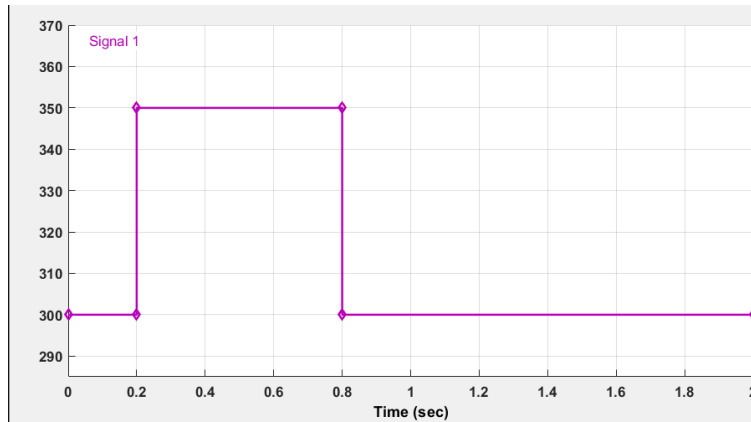


Figure (4) The input temperature of the Joule heating

As we see the system at 0 zero second the system should rise from 293 K to 300 K then at .2 second it will rise to 350 K and at .8 second it will drop again at 300 K

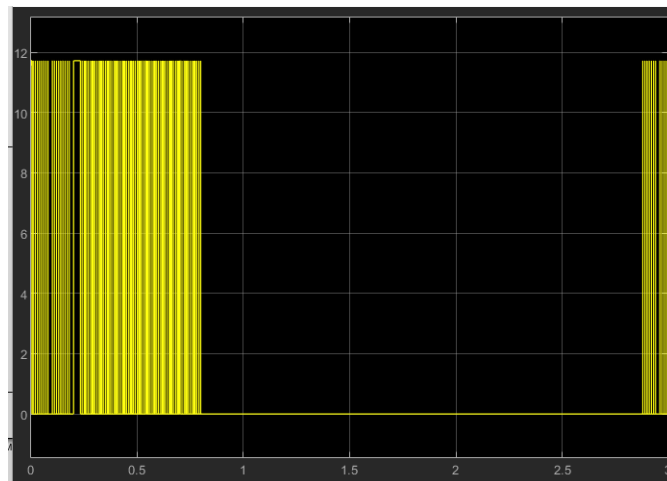


Figure (5) The PWM input Voltage

We see the signal is got denser to reach 350 K then it got to zero again to cool down to reach 300 K and get dense to maintain it .

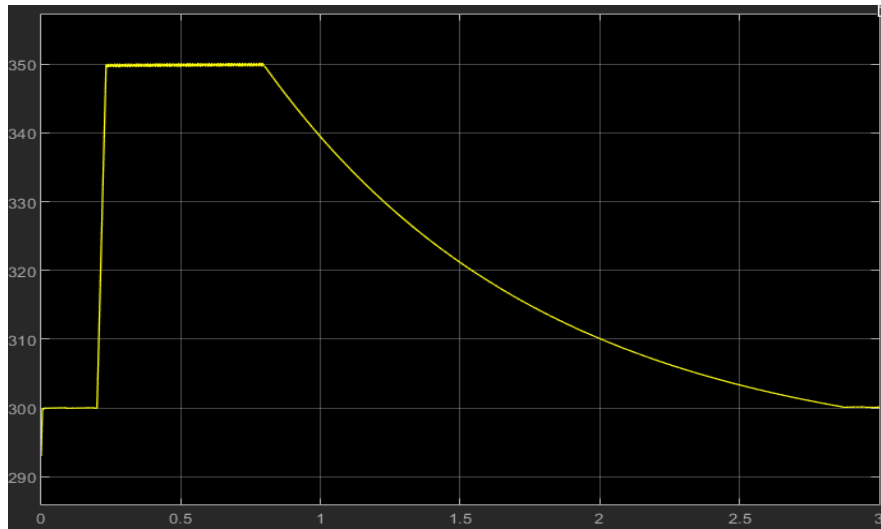


Figure (6) The output temperature

We see how the temperature rise to 300 K from 0 to .05 second and how it is reflected in Figure (4) a dense PWM is applied then the PWM becomes less dense to maintain it , becomes denser again to reach 350 K and it doesn't get less dense to maintain such a high temperature then the temperature went down to reach 300 K there is a 2 seconds delay then to maintain this temperature the PWM returns again . Next we 'll multiply the convection gain (h) by 3 and see how it affects the system ,it is also meaning in real life that I changes the cooling method

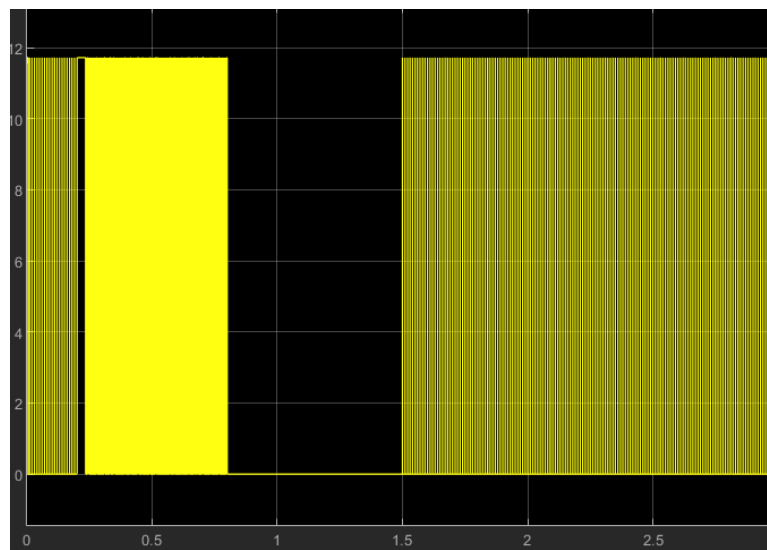


Figure (7) The second input PWM voltage

We see here the PWM is fighting harder to reach and maintain the temperature and the delay gets cut to .7 second

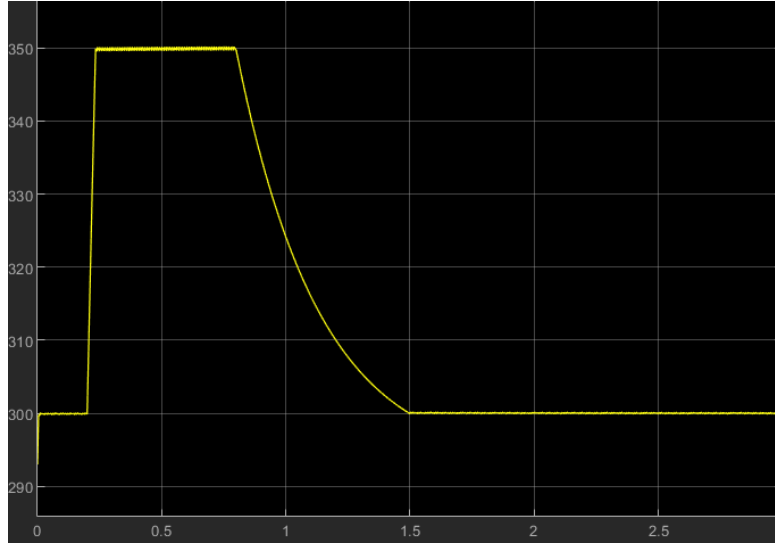


Figure (8) The output temperature of the second experiment

Figure (7) is the reflection of the previous PWM plot as the cutoff makes the system cool down then the duty cycle increases again to maintain the temperature at 300 K

4. Integrating Joule heating to SMA bending wire system

After testing how the Joule heating works as standalone let us take a step further and integrate it to the main system to get closer to real life simulation. Before integrating the joule heating to the system there is something to add to heat transfer equation. it is the latent heat it slows down both heat and cooling steps ΔH is Latent heat of transformation enthalpy it equals $24,000 \frac{J}{KG}$ and $\frac{d\xi}{dt}$ change of rate of martensite volume fraction

$$\frac{dT}{dt} = \frac{1}{m \cdot C_n} \left[\frac{V^2}{R} - hA_s(T - T_\infty) + m\Delta H \frac{d\xi}{dt} \right]$$

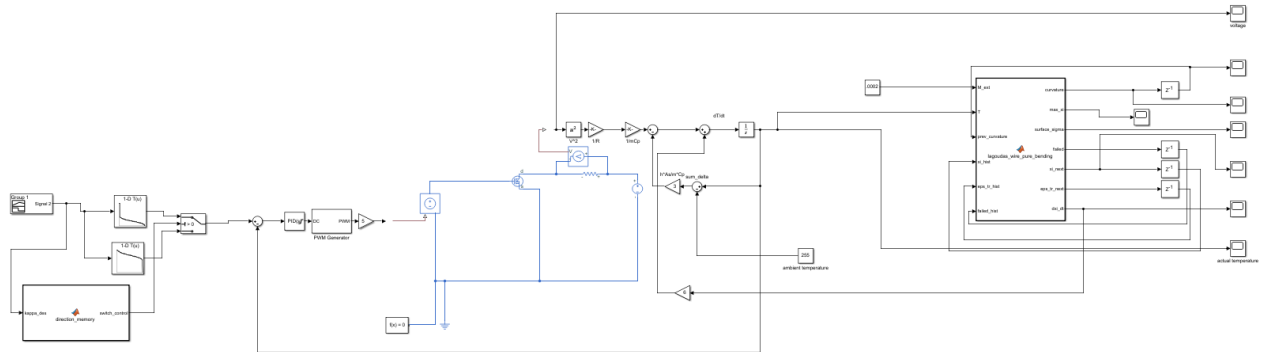


Figure (9) The Joule heating integrated SMA Bending wire

I preferred to not hide any part in subsystems to make it clearer to be seen how the system operates, first my input position signal goes the lookup tables configuration, it sends the targeted temperature to the Joule heating setup then the Joule heating starts to heat the wire, also I changed the ambient temperature (room temperature) to 255 K and added the latent heat I spoke about earlier in the documentation.

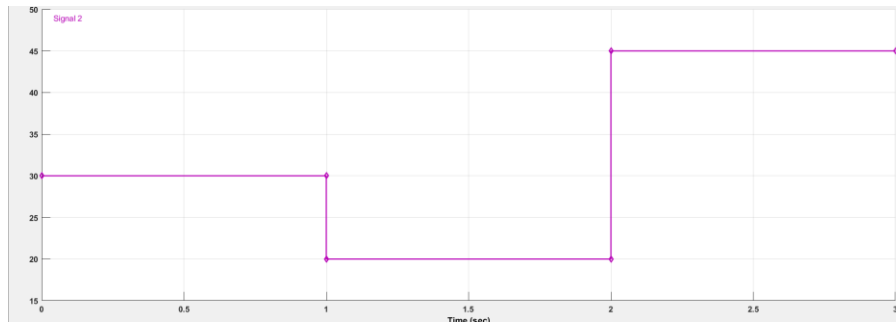


Figure (10) The input desired curvature trajectory signal



Figure (11) The input PWM voltage

Although the plot doesn't show clearly which is denser PWM area but we see clearly where the cooling down step and .6 second delay

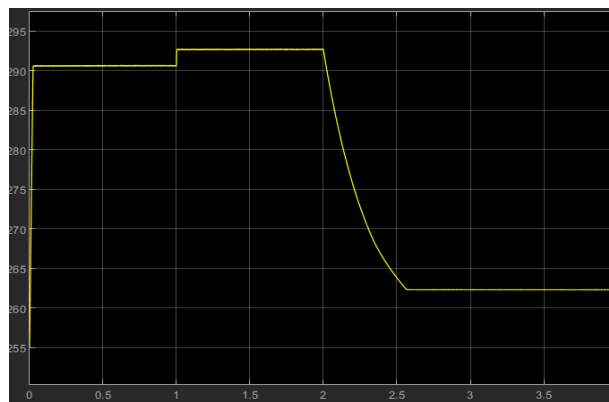


Figure (12) The applied temperature on the SMA wire

It shows here the gradual cooling to reach the final destination at 263 K.

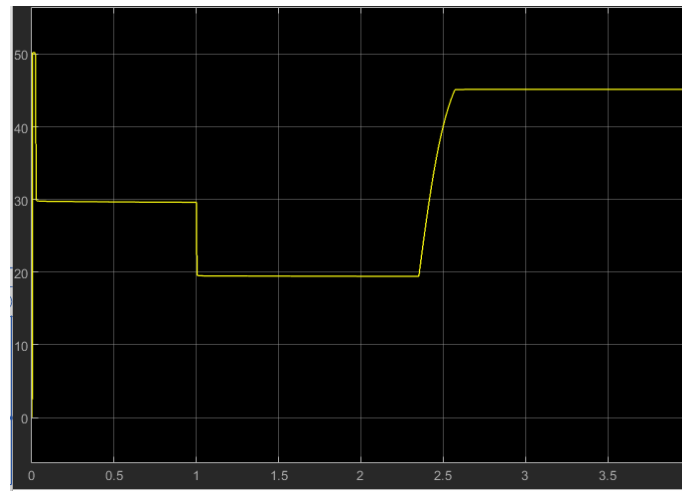


Figure (13) The output curvature

We see the bending load has its effects and bends the wire to $50\ m^{-1}$ curvature, the wire goes to $30\ m^{-1}$, continues its path to $20\ m^{-1}$ then reaches its final destination at $45\ m^{-1}$ with .6 second delay and .35 second dead band zone

5. The SMA wire safety and the balance of convection method and the applied voltage

If you notice the system was operating at 12 V, in the practical work that would be too dangerous and could cause permanent damage to the wire crystals and make it lose its shape memory effect, however if the electric power is lost due to the decline of the applied voltage the cooling gain also should be decreasing or it would overpower the electric power, in this section the system will operate at 3V with cooling gain 1 (The convective heat transfer coefficient (h) = $515\ \frac{W}{m^2 \cdot K}$), so we can't risk the wire for the sake of faster response

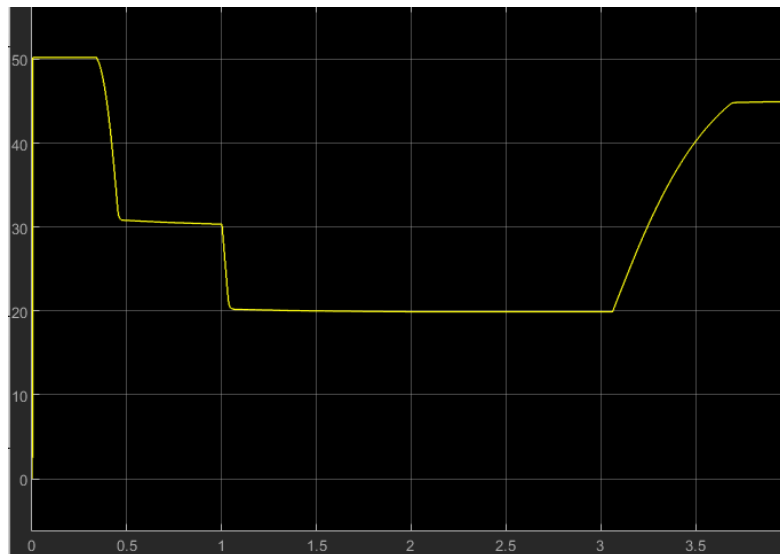


Figure (14) The output curvature

As we see here there is a .5 second delay in the first step these can be handled in the real practice by making the wire go right before the austenite starting phase temperature without taking in consideration it is given a command signal or not, now we should balance it more with a proper cooling method that would decrease the cooling step delay from 1.6 second.

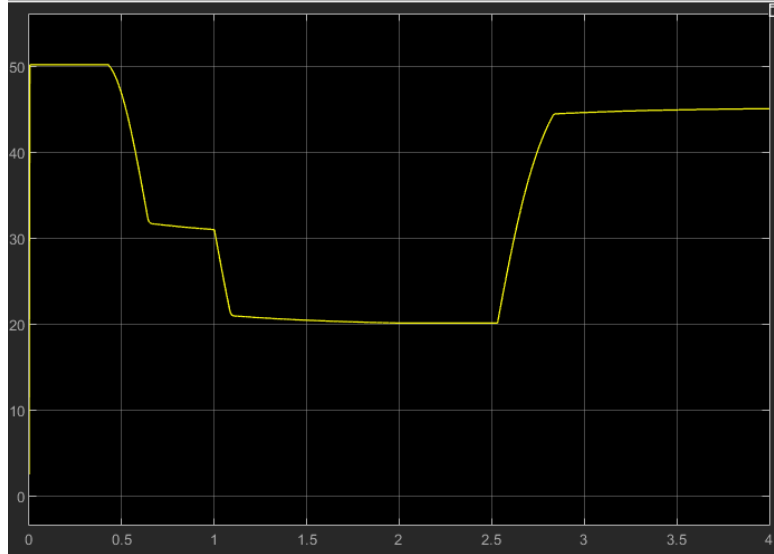


Figure (15) the output curvature with convection gain doubled

By comparing the Figure (14) and Figure (15), we notice that there is a tracking error the wire gets close to the destination at .6 seconds but doesn't go to 30 m^{-1} till 1 second, it happens again at 20 m^{-1} step, that's a something can be dealt with by hybrid feedforward controller in the next documentation without trying to tune the inner loop temperature PI controller, it is not a physics problem it a PI tuning problem.

6. The conclusion

We discussed how the PWM work with Joule heating and show that the convection gain can affect the heating curve as the PWM fights more to reach the targeted temperature also to maintain it as the convection gain rises due to changing the cooling method ,we also took a step closer to the real representation of the real world SMA wire actuator with implementing the Joule heating part it doesn't have that significant effect during heating put its effect appears very well during heating and we have shown the we can suppress the delay with change the cooling method in real life ,also we discussed the safety matter of the SMA wire that we can't easily risk the wire with high voltage for the sake of the fast response and we should balance the cooling method with the applied voltage the remaining problem now is the varying of the bending load problem as well as the tracking error that happened later in this documentation , I 'll discuss the two remaining problems in a standalone documentation.